

1. The physical significance of the Gauss's law for magnetism ( $\nabla \cdot \vec{B} = 0$ ) is

- ~~(1)~~ The magnetic field is conservative
- (2) Magnetic monopoles do not exist
- (3) Magnetic fields are produced only by static electric charges.
- (4) The magnetic field is directly proportional to the enclosed electric current.

2. Arrange the following steps in a communication process in sequential order:

- (a) Modulation
- (b) Message signal
- (c) Reception
- (d) Demodulation
- (e) Transmission

a - b - c d - e

~~(1)~~ (a)  $\rightarrow$  (b)  $\rightarrow$  (c)  $\rightarrow$  (d)  $\rightarrow$  (e) -

(2) (b)  $\rightarrow$  (a)  $\rightarrow$  (e)  $\rightarrow$  (c)  $\rightarrow$  (d)

(3) (a)  $\rightarrow$  (c)  $\rightarrow$  (b)  $\rightarrow$  (e)  $\rightarrow$  (d)

(4) (c)  $\rightarrow$  (b)  $\rightarrow$  (a)  $\rightarrow$  (d)  $\rightarrow$  (e) -

3. Match the following:

List-I  
(Physical Quantity)

List-II  
(SI Unit)

- |                      |                                     |
|----------------------|-------------------------------------|
| (a) Electric charge  | (i) Ohm-meter ( $\Omega\text{-m}$ ) |
| (b) Electric current | (ii) Watt (W)                       |
| (c) Resistivity      | (iii) Coulomb (C)                   |
| (d) Power            | (iv) Ampere (A)                     |

Codes:

(1) a - ii, b - i, c - iii, d - iv

~~(2)~~ a - iii, b - iv, c - i, d - ii

(3) a - iii, b - ii, c - iv, d - i

(4) a - i, b - iv, c - ii, d - iii

SPACE FOR ROUGH WORK

4. <sup>H<sup>+</sup> atom</sup> An alpha particle enters into a uniform magnetic field in perpendicular direction. Then it moves

- (1) Perpendicular to magnetic field in a straight line •
- (2) Parallel to magnetic field
- ~~-(3) In a circular orbit •~~
- (4) Comes to rest

5. Match the following:

| List-I<br>(Radiation)     | List-II<br>Wavelength Range (in Å) |
|---------------------------|------------------------------------|
| (a) Ultraviolet radiation | (i) 0.01 Å to 1 Å                  |
| (b) X-rays                | (ii) 7000 Å to 20000 Å             |
| (c) Infrared radiation    | (iii) 1000 Å to 4000 Å             |
| (d) Gamma radiation       | (iv) 1 Å to 100 Å                  |

Codes:

- (1) a - iii, b - iv, c - i, d - ii
- (2) a - ii, b - i, c - iv, d - iii
- (3) a - iv, b - iii, c - i, d - ii
- ~~(4) a - iii, b - iv, c - ii, d - i~~

6. For a composite molecule to be Raman active,

- (1) It should possess a permanent dipole moment.
- (2) It should possess a temporary dipole moment.
- ~~(3) It should exhibit changes in polarizability.~~
- (4) It should be a linear molecule

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SPACE FOR ROUGH WORK

7. Gravitational potential energy

- (a) Increases with increase in height from the surface of the earth
- (b) Decreases with increase in height from the surface of the earth
- (c) Is maximum at infinite distance from the surface of the earth
- (d) Is zero at infinite distance from the surface of the earth.

Which of the above statements is/are correct?

- (1) (a), (b) and (c) ~~(2)~~ (b) only
- (3) (a), (b), (c) and (d) (4) (a), (c) and (d)

8. Which of the following are the methods regarding measuring the velocity of ultrasonic waves in a solid?

- (a) Pulse-echo method
- (b) Young's modulus method
- (c) Piezoelectric method
- (d) Ferroelectric method

Choose the correct option.

- (1) (a), (b), (c) and (d) ~~(3)~~ (c) and (d)
- ~~(3)~~ (a) and (c) • (4) (d) only

9. Statement I: The speed of a transverse wave in a stretched string varies directly with the square root of the mass per unit length of the wave.

Statement II: All the particles oscillate with same amplitude in a stationary wave

Pick the correct answer regarding the above statements

- (1) Statement I is incorrect but Statement II is correct
- ~~(2)~~ Both Statement I and Statement II are correct
- (3) Statement I is correct but Statement II is incorrect
- (4) Both Statement I and Statement II are incorrect

SPACE FOR ROUGH WORK

$$v \propto \sqrt{m}$$

✓10. Our body gives heat to the surroundings and receives heat from the surroundings by the process of

(1) Conduction

(2) Convection

~~X~~ Radiation

~~X~~ Both Convection and Radiation

✓11. Consider the following statements and choose the correct option:

Statement I: Photogrammetry is used in stereoscopic measurements of human body using X-rays

Statement II: Accuracy of contours and cross-sections depends on the accuracy of 3D models generation from stereophotographs ✓

(1) Both Statements I and II are correct, but Statement II is not related to Statement I

(2) Both Statements I and II are correct, and Statement II is related to Statement I

(3) Statement I is correct, but Statement II is incorrect

~~X~~ Statement I is incorrect, but Statement II is correct

12. Match the concepts in List - I with their explanations in List - II and choose the correct answer:

List - I

List - II

(a) Exposure station

(i) The flying path an aircraft takes while taking the photographs

(b) Flightline

(ii) It is a vertical line from exposure station indicating the direction of gravity

(c) Perspective centre

(iii) Location of aircraft in the air at the time of taking photograph

(d) Plumblines

(iv) The point of origin or termination of bundles of perspective light rays

Codes:

(1) a - i, b - ii, c - iii, d - iv

~~X~~(2) a - iv, b - i, c - iii, d - ii

(3) a - ii, b - iii, c - iv, d - i

~~X~~(4) a - iii, b - i, c - iv, d - ii

SPACE FOR ROUGH WORK

13. Assume a vertical photograph was taken at a flying height of 5000 m above sea level using a camera with a 152 mm focal length lens, determine the photo scale points A and B, while they lie at elevations of 1200 m and 1960 m.

- (1) 1 : 25,000 - SA  
1 : 20,000 - SB
- (2) 1 : 20,000 - SA  
1 : 25,000 - SB
- (3) 1 : 25,000 - SA  
1 : 30,000 - SB
- (4) 1 : 30,000 - SA  
1 : 25,000 - SB

14. A line AB appears measuring 12.70 cm on an aerial photograph and the corresponding line measures 2.54 cm on a map at a scale 1/50,000. The terrain has an average elevation of 200 m above MSL. Calculate the flying height of aircraft, above MSL, if the photograph was taken with camera of 16 cm focal length.

- (1) 1600 m
- (2) 2000 m
- (3) 1800 m
- (4) 1400 m

15. The reference points, precisely located on both the ground and the photo whose three dimensional co-ordinates, are known as

- (1) Scale
- (2) Control points
- (3) Principle points
- (4) Photograph centre

16. Which of the following data sources can be used to generate a Digital Elevation Model (DEM)?

- (1) Topographic maps, stereo-air photos and stereo-satellite images
- (2) Only land survey data and GPS coordinates
- (3) Weather radar data and satellite thermal images
- (4) Multispectral satellite images and Sonar data

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SPACE FOR ROUGH WORK

17. Remote sensing is best described as
- (1) Acquiring information about any object without physical contact
  - (2) The measurement of celestial bodies
  - (3) Study of atmospheric pressure
  - (4) The measurement of soil characteristics

18. Consider the following statements and choose the correct option:

Statement I: Electromagnetic radiation is a form of energy, that travels with a speed of light ( $3 \times 10^8$  m/sec) in a harmonic pattern

Statement II: It has two fields: (i) Electric field and (ii) Magnetic field

- ~~(1)~~ Both Statements I and II are correct but Statement II is not related to Statement I ✓
- (2) Both Statements I and II are correct and Statement II is related to Statement I
  - (3) Statement I is incorrect, but Statement II is correct
  - (4) Statement I is correct, but Statement II is incorrect

19. The different stages of digital image processing in remote sensing are listed below

- (a) Image enhancement
- (b) Image preprocessing
- (c) Image classification
- (d) Image transformation

Identify the correct sequence.

- (1) (a), (b), (c), (d)
- (2) (b), (a), (d), (c)
- (3) (d), (b), (a), (c)
- (4) (c), (a), (b), (d)

20. Which of the following is true regarding the Indian Regional Navigation Satellite System (IRNSS)?

- (1) It consists of three satellites
- (2) It is a global positioning system
- (3) It was developed by the Department of Defence, USA
- ~~(4)~~ It is also known as NavIC and provides navigation signals over India and surrounding regions

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SPACE FOR ROUGH WORK

21. Which of the following problems is resolved by adding a 4<sup>th</sup> satellite observation to the minimum 3-range measurements needed for location tracking in GPS?
- ☒ (1) To calculate the exact velocity of the moving observer
  - (2) To measure the undulating shape of the earth
  - (3) To solve for the clock synchronisation error between the receiver and the satellite
  - (4) To boost the weak satellite signals so that they can work indoors
22. What does synchronisation do in a GIS database?
- (1) It speeds up computer internet connection
  - (2) It refers to conversion of vector data models to raster grid automatically
  - ☒ (3) It refers to forms of protection against inconsistencies that can result from multiple simultaneous users
  - (4) To build public web application without needing custom code
23. What is the correct sequence of the 4 M's in a standard GIS framework?
- (a) Mapping
  - (b) Monitoring
  - (c) Modelling
  - (d) Measurement
- Geographical Info System*
- ☒ (1) (a), (b), (c), (d)
  - ☒ (2) (d), (a), (b), (c)
  - (3) (c), (a), (d), (b)
  - (4) (a), (c), (b), (d)

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SPACE FOR ROUGH WORK

24. Consider the following statements:

Statement - I : GNSS can be easily integrated with total station data and GIS

Statement - II : GNSS gives precise results in areas like forest, under a bridge, tunnel etc.

Which one of the following is correct in respect of the above statements?

- ☒ (1) Both Statement-I and Statement-II are correct and Statement-II is related to Statement-I
- (2) Both Statement-I and Statement-II are correct but Statement-II is not related to Statement-I
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Statement-I is correct but Statement-II is incorrect

25. Which of the following statements about the Continuously Operating Reference Stations (CORS)

Network is incorrect?

- (1) CORS continuously receives signals from GNSS stations
- (2) Karnataka CORS network provides correction data for accurate positioning and surveying ✓
- (3) CORS operate 24/7 and serve as reference points for correcting GNSS measurements ✓
- ☒ (4) CORS data does not support GNSS surveying.

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SPACE FOR ROUGH WORK

Global Navigation Satellite Sys.

26. Match the following Dilution of Precision (DoP) in List-I with the Satellite Geometry in List-II:

## List - I

- (a) HDoP
- (b) PDoP
- (c) Low DoP
- (d) High DoP

## List - II

- (i) Better satellite geometry
- (ii) Poor satellite geometry
- (iii) Accuracy of horizontal position
- (iv) Position accuracy in 3-dimension

Codes:

- (1) a - iii, b - iv, c - i, d - ii
- (2) a - iii, b - i, c - iv, d - ii
- (3) a - i, b - ii, c - iii, d - iv
- (4) a - i, b - iii, c - ii, d - iv

27. Selective Availability in GPS signal is a

- (1) Natural error
- ~~(2) Intentional error~~
- (3) Systematic error
- (4) Bias error

28. Arrange the following steps in the correct sequence for obtaining position using DGPS:

- (a) Correction data is transmitted to the rover.
- (b) Rover determines its position using satellite signals.
- (c) Base station is established at a known point
- (d) Base station calculates the GPS errors
- (e) Correct co-ordinates of the rovers are obtained

(1) (c), (b), (d), (a), (e)

~~(2) (c), (d), (a), (b), (e)~~

(3) (d), (c), (b), (a), (e)

(4) (b), (c), (a), (d), (e)

SPACE FOR ROUGH WORK

**29.** Match List-I with List-II with respect to Real Time Kinematic (RTK) surveying and select the correct answer using the codes given below:

| List - I            | List - II                                                                              |
|---------------------|----------------------------------------------------------------------------------------|
| (a) Base station    | (i) Receives correction data and determines the rover position                         |
| (b) Rover receiver  | (ii) Transmits corrections from the base station to the rover                          |
| (c) Data link       | (iii) A signal distortion that happens when satellite waves bounce off nearby surfaces |
| (d) Multipath error | (iv) GNSS receiver established over a known coordinate point.                          |

Codes:

- (1) a - i, b - iv, c - iii, d - ii
- (2) a - i, b - ii, c - iv, d - iii
- ~~(3)~~ a - iv, b - i, c - ii, d - iii
- (4) a - iv, b - ii, c - i, d - iii

**30.** Which of the following best describes a satellite constellation?

- (1) A single satellite placed in a fixed orbit
- ~~(2)~~ A group of satellites working together to provide continuous or improved coverage
- (3) A ground-based network of GNSS receivers
- (4) A satellite tracking and monitoring station

**31.** Sukri is the tributary of which river?

- ~~(1)~~ Mahanadi
- (2) Luni
- (3) Rushi Kulya
- (4) Ponnaiyar

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SPACE FOR ROUGH WORK

32. Identify the mountains of Tamil Nadu in the map.



- (1) (a) Nilgiri (b) Javadi (c) Palani (d) Shevaroy  
 (2) (a) Nilgiri (b) Javadi (c) Shevaroy (d) Palani  
 (3) (a) Javadi (b) Nilgiri (c) Shevaroy (d) Palani  
 (4) (a) Javadi (b) Nilgiri (c) Palani (d) Shevaroy

33. India is a peninsular country because

- (1) It is located in the southern hemisphere  
 (2) It lies entirely in the Northern hemisphere  
 (3) It is located and surrounded by three sides of water and one side of land  
 (4) It is the seventh largest country in the world

34. The primary purpose of a cadastral map is to show

- (1) Physical features of the Earth's surface  
 (2) Weather patterns and climate data  
 (3) Political boundaries of countries and states  
 (4) Land ownership and property parcels

SPACE FOR ROUGH WORK

35. Match the types of map in List-I with their uses mentioned in List-II:

| List - I<br>(Types of map) | List - II<br>(Uses)                                                        |
|----------------------------|----------------------------------------------------------------------------|
| (a) Physical map           | (i) Shows political boundaries                                             |
| (b) Political map          | (ii) Focuses on specific theme                                             |
| (c) Topographic map        | (iii) Depicts physical features                                            |
| (d) Thematic map           | (iv) Uses contour lines to illustrate the elevation and shape of the land. |

Codes:

~~(1)~~ a - iii, b - i, c - iv, d - ii

(2) a - iv, b - iii, c - ii, d - i

(3) a - i, b - ii, c - iii, d - iv

(4) a - ii, b - i, c - iv, d - iii

36. Match List-I with List-II and choose the correct answer:

| List - I<br>(Discontinuity) | List - II<br>(Separate layers) |
|-----------------------------|--------------------------------|
| (a) Conrad                  | (i) Sial and Sima              |
| (b) Mohorovicic             | (ii) Outer core and inner core |
| (c) Gutenberg               | (iii) Crust and mantle Zone    |
| (d) Lehman                  | (iv) Mantle zone and core      |

Codes:

(1) a - ii, b - iv, c - i, d - iii

~~(2)~~ a - i, b - iii, c - iv, d - ii

(3) a - iii, b - i, c - iv, d - ii

(4) a - iv, b - ii, c - iii, d - i

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SPACE FOR ROUGH WORK

37. Consider the following statements and select the correct answer using the codes given below:

Statement I: In India, the drainage basin of Brahmaputra is bigger than that of the Indus basin

Statement II: The Mahanadi drainage basin is bigger than that of the Krishna river basin

(1) Only Statement I is correct

(2) Only Statement II is correct

(3) Both Statements I and II are correct

(4) Both Statement I and Statement II are incorrect

38. Choose the correct option.

(1) ITCZ is found between  $20^{\circ}$  -  $25^{\circ}$  latitude

(2) ITCZ is found between  $10^{\circ}$  -  $15^{\circ}$  latitude

(3) ITCZ is found between  $25^{\circ}$  -  $30^{\circ}$  latitude

(4) ITCZ is found between  $30^{\circ}$  -  $35^{\circ}$  latitude

39. Which of the following pairs is correctly matched?

(1) Abu - Malwa Plateau

(2) Tura Range - Shillong Plateau

(3) Pachmarhi - Aravalli Range

(4) Mukurti - Telangana Plateau

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SPACE FOR ROUGH WORK

40. The average depth of oceans and average height of the lithosphere, respectively, is

- (1) 3800 m and 840 m
- (2) 11000 m and 8848 m *highest*
- (3) 5000 m and 2500 m ✓
- (4) 4000 m and 3000 m

41. What is the primary function of GIS software environment's integration, with a Rational Database Management System (RDBMS)?

- (1) Compiling custom python scripts for tool automation
- ✓ (2) Managing and linking spatial geometry with non-spatial attribute tables efficiently
- (3) Rendering high definition 3D cartographic map layouts
- (4) Directly capturing live GPS satellite telemetry signals

42. What are the fundamental limitations and characteristics of Raster v/s Vector data models in representing geographic phenomena?

- ✓ (1) Vector models struggle with topological verification, while raster models automatically enforce neighbour relations through pixel positions
- (2) Vector models treat space as continuous field whereas raster models use discrete, entity based co-ordinates.
- ✓ (3) Raster grids lack explicit topology, whereas topological vector models explicitly store these connection using nodes arcs and polygons
- (4) Raster models require less storage than vector models for high complex, rapidly changing continuous surfaces

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SPACE FOR ROUGH WORK

43.

Match the Raster and Vector plugins in List – I with their corresponding cartographic application

in List – II:

**List – I**

(a) **Vector line operations**  
(eg: buffering)

(b) **Raster grid/pixel**  
**processing**

(c) **Vector to raster**  
**conversion**

(d) **Overlay analysis**  
(composite mapping)

**List – II**

(i) **Best suited for continuous surface backgrounds**

(ii) **Ideal for rendering precise land boundaries and**  
**transit networks with clean text labels**

(iii) **Combines multiple administrative and land cover**  
**layers to calculate cartographic overlap**

(iv) **Converting crisp outline geometries into pixel grids**  
**often causing jagged edges or loss in line resolution**

**Codes:**

(1) a – ii, b – i, c – iv, d – iii

(2) a – ii, b – i, c – iii, d – iv

~~(3) a – iv, b – iii, c – ii, d – i~~

(4) a – iii, b – ii, c – i, d – iv

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**SPACE FOR ROUGH WORK**

44.

Match the following OGC web services:

## List-I

## List-II

- |                      |                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------|
| (a) WMS              | (i) Returns a raw multidimensional raster data suitable for analysis                    |
| (b) WFS              | (ii) Core operation used to fetch the actual spatial features with geometric attributes |
| (c) WCS              | (iii) Returns styled map images (PNG, JPEG)                                             |
| (d) Get capabilities | (iv) Standard operation across all services                                             |
| (e) Get Feature      | (v) Returns raw vector features                                                         |

Codes:

~~1~~ a - iii, b - v, c - i, d - iv, e - ii

(2) a - ii, b - iv, c - v, d - i, e - iii

(3) a - i, b - ii, c - iii, d - iv, e - v

(4) a - v, b - iii, c - iv, d - ii, e - i

~~45.~~ What is the primary purpose of cadastral map within GIS framework?

(1) To analyse global weather and real time precipitation patterns

~~(2)~~ To define, record and legalise property boundaries

(3) To map out sub-surface geological rock formations

(4) To illustrate national political boundaries

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SPACE FOR ROUGH WORK

46. Given below are two statements related to AutoCAD software:

Statement I: Orthographic drawing refers to a 2D [2-Dimensional] representation of an object in a view that shows only one side at a time

Statement II: The file extension that cannot be accessed by AutoCAD is doc file

With respect to the above statements, choose the correct answer from the options given below:

- (1) Both Statements I and II are false
- ☒ (2) Both Statements I and II are true
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

47. Which of the following statements is/are not correct with respect to mirror command in AutoCAD?

- (a) Applies a mirror like reflection to an object
- (b) Creates a duplicate copy of an object
- (c) Flips an object over a specified axis
- (d) Rotates an object around its center point

- (1) (a) and (c)
- ☒ (2) (c) only
- (3) (a), (b) and (d)
- (4) (a), (b), (c) and (d)

48. Match the following function keys in List-I with their functions in List-II regarding to AutoCAD software:

| List-I | List-II                                                                           |
|--------|-----------------------------------------------------------------------------------|
| (a) F6 | (i) Toggles the SNAP on and off                                                   |
| (b) F7 | (ii) Toggles ORTHO on and off                                                     |
| (c) F8 | (iii) Toggles GRID on and off                                                     |
| (d) F9 | (iv) Toggles the co-ordinate readout from absolute to incremental to off and back |

Choose the correct option using the codes given below:

- (1) a - iv, b - iii, c - ii, d - i
- (2) a - ii, b - i, c - iv, d - iii
- (3) a - iii, b - iv, c - i, d - ii
- (4) a - iv, b - iii, c - i, d - ii

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SPACE FOR ROUGH WORK

49. Identify the correct file extension which is an open vector format primarily used for sharing CAD design across platforms, developed by Autodesk.

(1) .doc

(2) .dwg

(3) .dwt

~~(4) .dxf~~

50. The command for opening 'drawing units' window in AutoCAD is

(1) U

~~(2) UN~~

(3) UNI

(4) UIT

51. What will be the output of the following Java code?

```
class incre {
    public static void main(String args[ ])
    {
        int g = 3;
        system.out.print(++g * 8);
    }
}
```

(1) 33

(2) 34

(3) 24

~~(4) 32~~

52. Which line of the following HTML file contains an error?

1. <applet ?
2. Width = 200, Height = 200
3. Code = Hello.class
4. <param
5. Name = "string"
6. Value = "Hello java" >
7. </applet>

(1) Line 4

(2) Line 6

(3) Line 3

~~(4) Line 2~~

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SPACE FOR ROUGH WORK

53. How do you correctly declare a variable in PHP?

- (1) variable x = 5;
- (2) int x = 5;
- (3) \$x = 5; ✓
- (4) var x = 5; ✓

54. In a capacity based region quadtree used for point data, what primary condition triggers a leaf node to recursively split into four quadrants?

- ☒ (1) The number of data points inside the node exceeds a predefined capacity threshold
- (2) The depth of the entire tree decreases below zero
- (3) The bounding box of the node intersects with another branch of the tree
- (4) The node becomes completely empty of any spatial data

55. What is the defining characteristic of an "offline-first" mobile application architecture compared to an "online-first" cloud dependent application?

- (1) No local database on the device
- (2) Cannot function without an internet connection
- (3) No optimistic UI
- ☒ (4) It treats local device storage as the primary source of truth, rather than relying on a remote cloud server

56. What is the major limitation of the tradition waterfall model in Software Engineering?

- (1) It promotes iterative delivery, every two weeks
- (2) It eliminates the need of formal testing phase
- ☒ (3) Requirements are freezed
- (4) Requires continuous collaboration

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SPACE FOR ROUGH WORK

57. Consider the following two statements related to REST API.

Statement I: Stateless constraint of REST requires that each request from a client to server must contain all information necessary to understand and complete request

Statement II: Because API server stores user session data in its local <sup>cache</sup> memory to track state across successive requests, the server is forced to remember previous interactions, which directly violates this constraint

With respect to the above statements, choose an appropriate answer from the options given below:

☒ Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

58. What is the correct sequence of software testing levels from lowest component level to complete system level?

(a) System Testing

(b) Integration Testing

(c) Unit Testing

(d) Acceptance Testing

Choose the correct sequence from the below mentioned options.

(1) (d), (a), (b), (c)

☒ (2) (c), (b), (a), (d)

(3) (c), (a), (b), (d)

(4) (d), (c), (a), (b)

SPACE FOR ROUGH WORK

59. Choose the correct sequential order of Jesse James Garrett's 'Five Planes Framework' in UX design:

- |               |             |
|---------------|-------------|
| (a) Strategy  | (b) Scope   |
| (c) Structure | (d) Surface |
| (e) Skeleton  |             |

- (1) (a), (b), (d), (e), (c)  
 (2) (a), (b), (c), (e), (d)  
 (3) (b), (d), (a), (e), (c)  
 (4) (e), (d), (b), (a), (c)

60. Which type of cryptographic mechanism serves as technological foundation for standard secure digital signatures to ensure non-repudiation?

- (1) Shared secret key exchange  
 (2) Hash based message authentication codes  
~~(3) Asymmetric key cryptography (public key infrastructure)~~  
 (4) Symmetric key cryptography

61. Arrange the following steps in a proper sequence to construct an equilateral triangle of side 6 cm:

- (a) Join A and C  
 (b) Take a radius of 6 cm in a compass, draw an arc from A above the segment and cut this arc from B using the same radius. Mark the intersection of these arcs as C.  
 (c) Draw a segment AB of length 6 cm.  
 (d) Join B and C.

- (1) (a), (b), (c), (d)                      ~~(2)~~ (c), (b), (a), (d)  
 (3) (a), (c), (b), (d)                      (4) (b), (d), (c), (a)

---

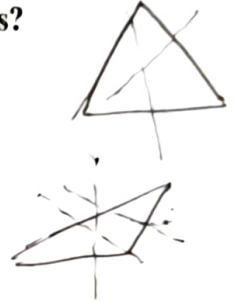
SPACE FOR ROUGH WORK

62. Statement I: Perpendicular bisectors of a triangle passes through the vertices of the triangle.

Statement II: Perpendicular bisectors of a triangle passes through the midpoints of sides of the triangle.

Which of the following conclusions can be drawn from the above statements?

- (1) Both the Statements I and II are true  
~~(2) Both the Statements I and II are false~~  
 (3) Statement I is true but Statement II is false  
 (4) Statement I is false but Statement II is true



63. In  $\triangle ABC$ ,  $AB = AC = x$ ,  $BC = 10$  cm and area of  $\triangle ABC = 60$   $\text{cm}^2$ . Then the value of  $x$  is

- (1) 12 cm  
 (2) 5 cm  
 (3) 10 cm  
~~(4) 13 cm~~



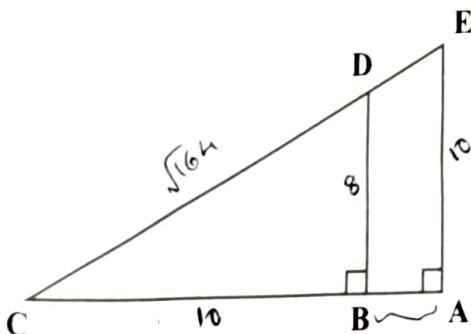
$$\frac{1}{2}(x+5)(5)(x-5) = 60$$

$$8 \sqrt{x^2 - 5^2} = 12$$

$$\frac{2x+10}{2} = 5$$

$$x =$$

64. If  $AE = 10$  cm,  $BD = 8$  cm and  $BC = 10$  cm, then  $AB$  is equal to



$$\sqrt{164}$$

$$\begin{array}{r} 2 \overline{) 164} \\ 2 \overline{) 82} \\ 41 \end{array}$$

$$2\sqrt{41}$$

$$\begin{array}{l} 4 \frac{8}{10} = \frac{10}{x} \\ 5 \\ x = \frac{50}{4} \\ 12.5 \end{array}$$

- (1) 5 cm  
 (2) 25 cm  
~~(3) 12.5 cm~~  
 (4) 2.5 cm

SPACE FOR ROUGH WORK

65. Match the following List-I with List-II:

List - I

- (a) Opposite sides of a parallelogram
- (b) The diagonals of a parallelogram
- (c) In parallelogram, consecutive angles are

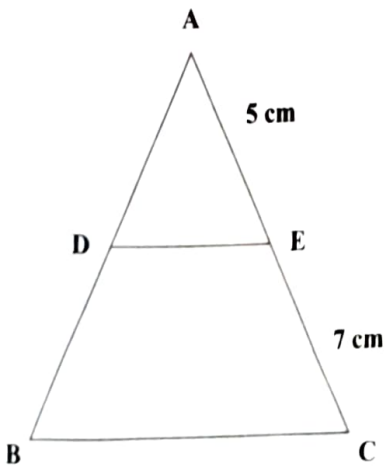
List - II

- (i) Supplementary angle
- (ii) Equal
- (iii) Parallelogram
- (iv) Bisect each other

Codes:

- ~~(1)~~ a - ii, b - iv, c - i
- (2) a - ii, b - iii, c - iv
- (3) a - iii, b - ii, c - i
- (4) a - iv, b - i, c - ii

66. In the following figure, BC is parallel to DE. Then the ratio, Area of  $\triangle ADE$  : Area of  $\triangle ABC$  is \_\_\_\_\_ (in  $\text{cm}^2$ )



(1) 25 : 49

(2) 49 : 25

~~(3)~~ 25 : 144

(4) 144 : 25

SPACE FOR ROUGH WORK

67. List-I denotes the polygons and List-II denotes the number of sides of polygons.

Match the following:

| List - I<br>(Polygons) | List - II<br>(Number of sides) |
|------------------------|--------------------------------|
| (a) Decagon            | (i) 8                          |
| (b) Hexagon            | (ii) 5                         |
| (c) Octagon            | (iii) 10                       |
| (d) Pentagon           | (iv) 6                         |

Codes:

(1) a - iv, b - iii, c - ii, d - i

(2) a - ii, b - iii, c - iv, d - i

~~(3) a - iii, b - iv, c - i, d - ii~~

(4) a - ii, b - i, c - iv, d - iii

68. The earning of a person is Rs. 40,000 in the first year and he earns Rs. 3000 more every year

for the next 15 years. Find the total amount he has earned in 16 years.

(1) Rs. 8,00,000

$$a = 40$$

$$d = 3$$

$$S_{16} = 8 \times [80 + 15 \times 3]$$

$$= 125 \times 8$$

$$1000,000$$

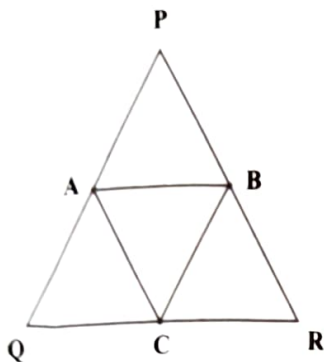
~~(2) Rs. 10,00,000~~

(3) Rs. 10,30,000

(4) Rs. 8,30,000

SPACE FOR ROUGH WORK

69. In the following figure, A, B and C are midpoints of PQ, PR and QR respectively. If the perimeter of  $\Delta PQR$  is 36 cm, then the perimeter of  $\Delta ABC$  is



- (1) 9 cm  
~~(2) 18 cm~~  
 (3) 20 cm  
 (4) 36 cm
70. Match each of the set in List-I described in the roster form with the same set in List-II described in the set builder form:

## List-I

- (a)  $\{1, 2, 3, 6, 9, 18\}$   
 (b)  $\{1, 2, 3, 6\}$   
 (c)  $\{4, -4\}$   
 (d)  $\{1, 3, 5, 7, 9\}$

## List-II

- (i)  $\{x : x \text{ is an integer and } x^2 - 16 = 0\}$   
 (ii)  $\{x : x \text{ is an odd natural number less than } 10\}$   
 (iii)  $\{x : x \text{ is a natural number and divisor of } 6\}$   
 (iv)  $\{x : x \text{ is a positive integer and a divisor of } 18\}$

Codes:

~~(1)~~ a - iv, b - iii, c - i, d - ii

(2) a - iv, b - ii, c - i, d - iii

(3) a - iii, b - i, c - ii, d - iv

(4) a - iii, b - iv, c - ii, d - i

SPACE FOR ROUGH WORK

$$(x+y+z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$$

$$(2+3)^2 = 25$$

D1

71. If  $x + y + z = 8$  and  $xy + yz + zx = 12$ , then the value of  $x^2 + y^2 + z^2$  is

~~(1) 40~~

(2) - 40

(3) 38

(4) - 38

$$a^4 + a^2b^2 - 2ab^3 + b^4$$

72. Factorize:  $a^4 + a^2b^2 + b^4$

(1)  $(a^2 - b^2 + ab)(a^2 - b^2 - ab)$

(2)  $(a^2 - b^2 + ab)(a^2 + b^2 - ab)$

(3)  $(a^2 + b^2 + ab)(a^2 - b^2 - ab)$

~~(4)  $(a^2 + b^2 + ab)(a^2 + b^2 - ab)$~~

73. Statement I: If  ${}^nC_8 = {}^nC_2$ , then  ${}^nC_2 = 45$  ✓

$$\frac{5 \times 4}{2} = 10$$

Statement II: If  ${}^nC_x = {}^nC_y$ , then  $n = x + y$

(1) Statement I is true but Statement II is false

(2) Statement II is true but statement I is false

~~(3) Both Statements I and II are true~~

(4) Both Statements I and II are false

74. The difference between two numbers is 2 and the product of their LCM and HCF is 48.

The numbers are

$$(a-b) = 2$$

~~(1) 8 and 6~~

(2) 11 and 9

$$ab = 48$$

(3) 10 and 12

(4) 6 and 4

$$(a+b)b = 48$$

$$b^2 + 2b - 48 = 0$$

75. Mode is equal to

$$b^2 + 2b - 48$$

(1) 3 median + 3 mean

$$b(b+8) - 6(b+8)$$

(2) 3 median + 2 mean

$$6, -8$$

(3) 2 median - 3 mean

~~(4) 3 median - 2 mean~~

SPACE FOR ROUGH WORK

76. If  $A = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 0 & 1 \\ 5 & 2 & 6 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 1 & 4 \\ 2 & 6 & 1 \end{bmatrix}$  and  $2A + 3X = 5B$ , then what is  $X$ ?

(1)  $\frac{1}{3} \begin{bmatrix} 8 & 1 & 9 \\ 2 & 0 & 18 \\ 0 & 20 & -7 \end{bmatrix}$

~~(2)~~  $\frac{1}{3} \begin{bmatrix} 8 & -1 & 9 \\ 2 & 5 & 18 \\ 0 & 20 & -7 \end{bmatrix}$

(3)  $\frac{1}{3} \begin{bmatrix} 8 & -1 & 9 \\ 2 & 5 & 18 \\ 0 & 20 & 7 \end{bmatrix}$

~~(4)~~  $\frac{1}{3} \begin{bmatrix} 8 & -1 & -9 \\ 2 & 5 & 18 \\ 0 & 20 & -7 \end{bmatrix}$

Handwritten calculations for Q76:  
 $2A = \begin{bmatrix} 2 & -4 & 6 \\ -2 & 0 & 2 \\ 10 & 4 & 12 \end{bmatrix}$   
 $5B = \begin{bmatrix} 10 & -5 & 15 \\ 0 & 5 & 20 \\ 10 & 30 & 5 \end{bmatrix}$   
 $3X = 5B - 2A = \begin{bmatrix} 8 & -1 & 9 \\ 2 & 5 & 18 \\ 0 & 20 & -7 \end{bmatrix}$   
 $X = \frac{1}{3} \begin{bmatrix} 8 & -1 & 9 \\ 2 & 5 & 18 \\ 0 & 20 & -7 \end{bmatrix}$

77. The distance between two parallel tangents of a circle of radius 6 cm is \_\_\_\_\_

(1) 10 cm

~~(2)~~ 12 cm

(3) 8 cm

(4) 6 cm

Handwritten calculations for Q77:  
 $3 \overline{) 12152130}$   
 $\underline{45710}$

78. The largest four-digit number which is divisible by 12, 15, 21 and 30 is

(1) 9340

(2) 9831

(3) 9960

~~(4)~~ 9660

Handwritten calculations for Q78:  
 $60 \times 7 \times 10 = 4200$   
 $4200 \times 2 = 8400$

79. The abscissa of a point is  $\frac{3}{2}$  units, and if the difference between the ordinate and  $\frac{5}{2}$  is double of the abscissa, then the value of the ordinate is

(1)  $\frac{3}{2}$

(2)  $\frac{5}{2}$

(3)  $\frac{7}{2}$

(4)  $\frac{11}{2}$

SPACE FOR ROUGH WORK

80. In a parallelogram, the consecutive angles are

- (1) Equal angles (2) Complementary angles  
(3) Right angles (4) Supplementary angles

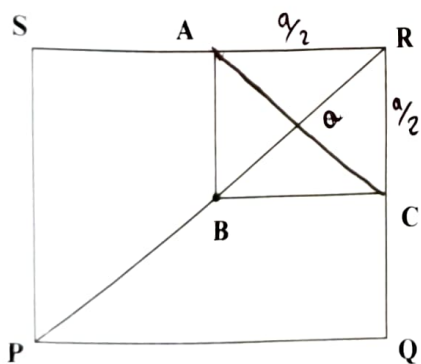
81. Match the following List-I the name of the matrix with List-II examples of matrices.

| List-I              | List-II                                              |
|---------------------|------------------------------------------------------|
| (a) Identity matrix | (i) $[a \ 0 \ a]$                                    |
| (b) Diagonal matrix | (ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  |
| (c) Column matrix   | (iii) $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ |
| (d) Row matrix      | (iv) $\begin{bmatrix} a \\ b \end{bmatrix}$          |

Codes:

- (1) a - ii, b - iii, c - iv, d - i (2) a - iii, b - ii, c - i, d - iv  
(3) a - iv, b - ii, c - iii, d - i (4) a - i, b - iv, c - ii, d - iii

82. In the following figure, PQRS and ABCR are the rectangles and B is the midpoint of PR. The ratio of lengths AC : PR is \_\_\_\_\_



Handwritten calculations:

$$a\sqrt{2}$$

$$\sqrt{\frac{a^2}{2} + \frac{a^2}{2}} = a\sqrt{2}$$

$$\frac{\sqrt{2a^2}}{\sqrt{2}} = \frac{a}{\sqrt{2}}$$

$$\frac{a}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$$

- (1) 1 : 2 (2) 2 : 1  
(3) 1 : 1 (4) 2 : 3

SPACE FOR ROUGH WORK

83. Match the following square roots and cube roots in List-I with their corresponding answers in List-II.

| List-I                                      | List-II |
|---------------------------------------------|---------|
| (a) $\sqrt[3]{2744}$                        | (i) 8   |
| (b) $\left(\frac{2}{3}\right)^3 \sqrt{729}$ | (ii) 30 |
| (c) $\sqrt{18^2 + 24^2}$                    | (iii) 4 |
| (d) $\sqrt[3]{4096}$                        | (iv) 14 |

$$\begin{array}{l}
 9^3 \quad \frac{81 \times 9}{729} \\
 2^3 \times 3^3 = 8 \times 27 = 216 \\
 8 \times 9 = 72
 \end{array}$$

Codes:

(1) a - ii, b - iii, c - iv, d - i

(2) a - iv, b - iii, c - i, d - ii

(3) a - iii, b - iv, c - ii, d - i

~~(4)~~ a - iv, b - i, c - ii, d - iii

84. The area of a trapezium is  $144 \text{ m}^2$  and its height is 12 m. If one of the parallel sides is 14 m less than the other, then find the length of each of the parallel sides.

$$\begin{array}{l}
 \frac{1}{2} (a+b) \times h \\
 \frac{1}{2} (a+b) \\
 a + b = 24
 \end{array}$$

~~(1)~~ 19 m and 5 m

(2) 20 m and 4 m

(3) 19 m and 4 m

(4) 20 m and 5 m

85. The difference between the number of bases in a prism and a pyramid is \_\_\_\_\_

(1) A prism has one base and a pyramid has two bases

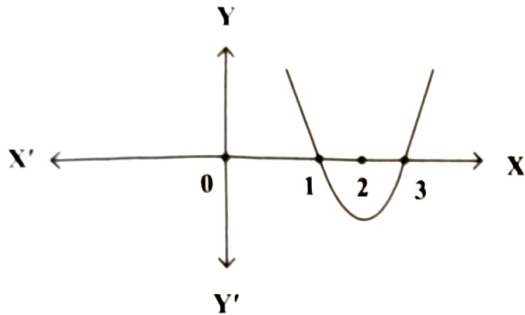
~~(2)~~ A prism has two parallel bases and a pyramid has one base

(3) Both have the same number of bases

☐ (4) A prism has no base

SPACE FOR ROUGH WORK

86. The quadratic equation of the given graph is



~~(1)~~  $x^2 - 4x + 3 = 0$

(2)  $x^2 - 4x - 3 = 0$

(3)  $x^2 + 4x - 3 = 0$

(4)  $x^2 + 4x + 3 = 0$

87. The base of a parallelogram is thrice of its height. If the area of the parallelogram is  $108 \text{ cm}^2$ , then its height is

(1) 8 cm

(2) 5 cm

~~(3)~~ 6 cm

(4) 4 cm

$$\begin{aligned} b \times h &= 108 & 3h^2 &= 108 \\ 3h &= 108 & h^2 &= 36 \\ 3h &= 27 & h &= 9 \end{aligned}$$

88. If the four angles of a quadrilateral are in the ratio  $2 : 3 : 5 : 8$ , find the angles.

~~(1)~~  $40^\circ, 60^\circ, 100^\circ$  and  $160^\circ$

(2)  $20^\circ, 80^\circ, 100^\circ$  and  $160^\circ$

(3)  $40^\circ, 80^\circ, 160^\circ$  and  $180^\circ$

(4)  $60^\circ, 80^\circ, 160^\circ$  and  $180^\circ$

$$\begin{aligned} 18x &= 360 \\ x &= 20 \end{aligned}$$

40

89.  $\triangle ABC$  has sides which are of length 5 cm, 12 cm and 13 cm respectively. The radius of circle circumscribing the triangle is

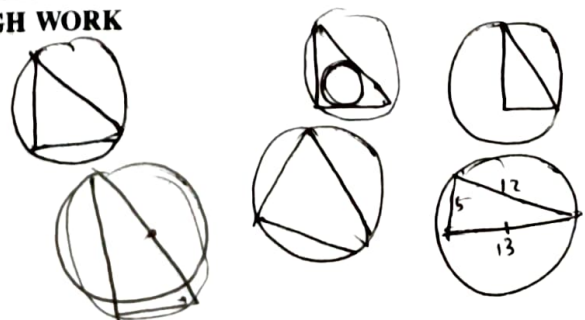
~~(1)~~ 5 cm

(2) 12 cm

(3) 2.5 cm

~~(4)~~ 6.5 cm

SPACE FOR ROUGH WORK



90. How many 4 - digit numbers can be formed by using the digits 1 to 9 if repetition of digits is not allowed?

(1) 3024

(2) 3034

(3) 4030

(4) 5060

$$\begin{array}{r} 9 \times 8 \times 7 \times 6 \\ \hline 3024 \end{array}$$

91. Match the following Algebraic expressions in List-I with their simplified form in List-II.

List-I

List-II

(a)  $(x - 3y)(x + 5y)$

(i)  $x^2 - 8xy + 15y^2$

(b)  $(x + 3y)(x + 5y)$

(ii)  $x^2 - 2xy - 15y^2$

(c)  $(x - 3y)(x - 5y)$

(iii)  $x^2 + 8xy + 15y^2$

(d)  $(x + 3y)(x - 5y)$

(iv)  $x^2 + 2xy - 15y^2$

$$x^2 + 2xy - 15y^2$$

$$\begin{array}{r} 7 \times 7 \times 6 \\ \hline 504 \times 6 \\ \hline 3024 \end{array}$$

Codes:

(1) a - iii, b - ii, c - iv, d - i

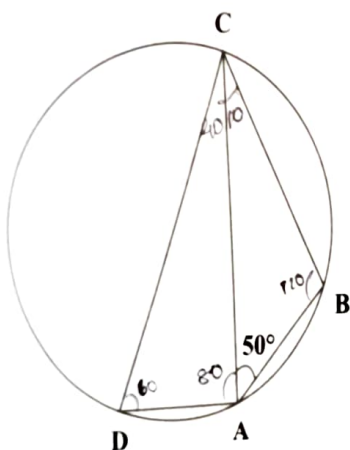
(2) a - iv, b - iii, c - i, d - ii

(3) a - ii, b - iv, c - iii, d - i

(4) a - ii, b - i, c - iii, d - iv

92. From the given figure, if  $\angle ADC : \angle ABC = 1 : 2$ , then what is the value of  $\angle ADC$ ? Choose the correct answer from the given options.

60°



$$\frac{x}{y} = \frac{1}{2}$$

$$x + y = 180^\circ$$

$$3x = 180^\circ$$

$$x = 60^\circ$$

$$x + 2x = 180$$

$$\begin{array}{r} 120 \\ 60 \end{array}$$

$$\begin{array}{r} +50 \\ 130 \\ 40 \end{array} = 180^\circ$$

(1) 20°

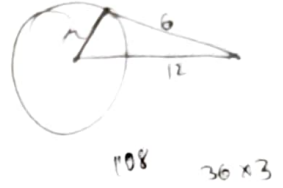
(2) 30°

(3) 80°

(4) 10°

SPACE FOR ROUGH WORK

93. The length of the tangent drawn from a point which is 12 cm away from the center of a circle is 6 cm. Find the radius of the circle.



(1) 6 cm

(2)  $2\sqrt{3}$  cm

~~(3)  $6\sqrt{3}$  cm~~

(4) 9 cm

94. The steps from solids to points are

(1) Lines - Points - Surfaces - Solids

(2) Solids - Surfaces - Lines - ~~(points)~~

(3) Lines - Surfaces - Points - Solids

~~(4) Solids - Surfaces - Lines - Points~~

95. Which of the following sets are infinite sets.

(a) The total population of India.

(b) Set of Natural numbers.

(c) Set of Prime numbers.

(d) Number of water drops in a glass.

(1) (a) and (c)

~~(2) (b) and (c)~~

(3) (a) and (d)

(4) (b), (c) and (d)

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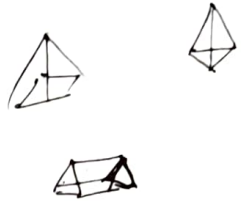
SPACE FOR ROUGH WORK

96. Which point of concurrency is equidistant from the three sides of a triangle and serves as the center of its inscribed circle?

- (1) Centroid
- ~~(2) Incenter~~
- (3) Orthocenter
- (4) Circumcenter

97. How many faces and edges does a triangular prism have respectively?

- (1) 8 and 9
- (2) 9 and 8
- (3) 9 and 5
- ~~(4) 5 and 9~~



98. Arrange the following polynomials in an ascending order of their highest degrees:

- (a)  $(x^2 + 3)(x^2 + 1)$  4
- (b)  $2x^2(x^3 + 4x + 1)$  5
- (c)  $x(2x^2 + 3x + 1)$  3
- (d)  $(x + 1)(3x + 2)$  2

- (1) (b)  $\rightarrow$  (a)  $\rightarrow$  (c)  $\rightarrow$  (d)
- (3) (b)  $\rightarrow$  (c)  $\rightarrow$  (a)  $\rightarrow$  (d)

- ~~(2)~~ (d)  $\rightarrow$  (c)  $\rightarrow$  (a)  $\rightarrow$  (b)
- (4) (d)  $\rightarrow$  (c)  $\rightarrow$  (b)  $\rightarrow$  (a)

99. 'A' and 'B' together can do a piece of work in 4 days. If one day's work of 'A' is two times one day's work of 'B', then how many days will each take to finish the work if 'A' and 'B' work individually?

- (1) A: 4 days and B: 3 days
- (2) A: 3 days and B: 2 days
- (3) A: 2 days and B: 4 days
- ~~(4) A: 6 days and B: 12 days~~

$$(E_A + E_B) \times 4 = W$$

$$3E_B \times 4 = W$$

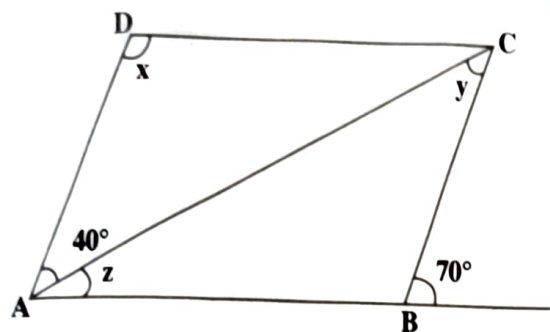
$$E_A = 2E_B$$

$$E_B = \frac{1}{12}$$

$$E_A = \frac{1}{6}$$

SPACE FOR ROUGH WORK

100. In the following figure, ABCD is a parallelogram. Find the angle measures of  $x$ ,  $y$  and  $z$ .



$$\begin{aligned} y &= 40^\circ \\ z &= 30^\circ \\ x &= 110^\circ \end{aligned}$$

- (1)  $\angle x = 110^\circ$ ,  $\angle y = 30^\circ$ ,  $\angle z = 40^\circ$   
~~(2)  $\angle x = 110^\circ$ ,  $\angle y = 40^\circ$ ,  $\angle z = 30^\circ$~~   
 (3)  $\angle x = 40^\circ$ ,  $\angle y = 110^\circ$ ,  $\angle z = 30^\circ$   
 (4)  $\angle x = 40^\circ$ ,  $\angle y = 30^\circ$ ,  $\angle z = 110^\circ$

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SPACE FOR ROUGH WORK